P525/1
CHEMISTRY
Paper 1
July,2026
2¾ hours

ASSHU ANKOLE JOINT MOCK EXAMINATIONS
Uganda Advanced Certificate of Education
CHEMISTRY
(Theory)
Paper 1

2 hours 45 minutes

INSTRUCTIONS TO CANDIDATES

*This examination paper consists of **two** sections **A** and **B**. It has **six** examination items.*
*Section A has two **compulsory** items.*

*Section **B** has two parts; **I** and **II**. Respond to only **one** item from each part.*

*Respond to **four** items **in all**.*

*Any additional item(s) responded to will **not** be scored.*

*All responses **must** be written in the answer booklet(s) provided.*

Begin each item on a fresh page.

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SECTION A

Respond to all items in this section

Item one

As the demand for clean and sustainable fuel continues to grow, an energy company in Tororo is exploring the production of methanol for use in direct methanol fuel cells and as a petrol additive. Methanol is regarded as a promising fuel because it burns cleanly and can be produced from natural gas, reducing Uganda's dependence on imported petroleum.

Before establishing a full-scale production plant, the company is assessing a two-step process that produces methanol from methane and steam. The company also wants to investigate the conditions that affect the rate and yield of methanol production. The company intends to establish production units in remote communities where efficiency, safety, and sustainability are critical.

The two-step reactions are represented by the equations:

Step1: $CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g)$
Step 2: $CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$

A series of experiments and data analysis was conducted at 500°C and the following results were obtained.

Table.1: Kinetic data for step 2.

Experiment number.	Initial [CO] (mol dm^{-3})	Initial[H_2] (mol dm^{-3})	Initial rate ($moldm^{-3}S^{-1}$)
1	0.010	0.010	0.005
2	0.020	0.015	0.030
3	0.010	0.030	0.015

Table 2: Thermochemical data.

Substance	$CH_4(g)$	$H_2O(g)$	$CO(g)$	$CH_3OH(g)$
Enthalpy of formation ($Kjmol^{-1}$)	-75	-242	-111	-201

The company chemist has advised that a rate constant above 4450 $mol^{-2}dm^6s^{-1}$ and an overall order greater than 2 in **step 2**represent an efficient production process.

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Task:

As a student chemist:

a) Analyse the data provided and use it to comment on whether the reaction can actually be used for electricity generation and help the company assess the efficiency of the production process.
b) Explain the conditions the company should optimise to increase the production rate of the fuel.
c) Analyse the importance of these laws in the scenario for real-life applications and make a recommendation regarding the qualities of a good fuel that the company should consider

Item two

A chemical company in Kasese district intends to produce sulphuric acid to support the manufacture of phosphate fertilizers for farmers in Uganda. Sulphuric acid is produced industrially using the Contact Process.

Before installing a full-scale plant, the company's laboratory team carried out two pilot experiments

Experiment 1

A 2.0 dm^3 reaction vessel was filled with 3.0 moles of sulfur dioxide gas and 2.0 moles of oxygen gas. The mixture was heated to 450 °C in presence of vanadium(V) catalyst. At equilibrium, 1.8 moles of sulfur trioxide had been formed according to the following equation:

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g) \quad -197 kjmol^{-1}$

Experiment 2

The Sulphur trioxide gas from Experiment 1 was absorbed in concentrated sulphuric acid to form oleum, which was then carefully diluted with water to form 98 % concentrated sulfuric acid.

For fertilizer processing, the company prepares a working solution by diluting 5.0cm^3 of 98% sulphuric acid in distilled water to make 500 cm^3 of dilute solution.

The company has been advised that an equilibrium constant, Kc value greater than 3.60 $mol^{-1}dm^3$ in sulphur trioxide synthesis would represent optimal production efficiency and the pH of the resultant diluted sulphuric acid solution should be below 0.54 for efficient fertilizer production. *(Density of sulphuric acid is 1.84gcm^{-3}, H=1, S=32, O=16)*

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The company also wants to investigate the conditions that affect the rate and yield of sulfuric acid production.

Tasks:

As a learner of chemistry;

a) Analyse the data provided and use it to comment on the production efficiency of the company.
b) Explain the ways the company would use to increase the amount of sulphuric acid produced during the production process.
c) Explain other real-life applications of the principles used in the company.

SECTION B

Part I

Respond to any one item in this part

Item 3

Transition metals are widely used in industry because of their unique properties. In Jinja, a leather tannery uses chromium compounds to treat animal hides. The water treatment and fertilizer plants use manganese compounds and iron in water purification and as catalyst in the Haber process respectively.

During a joint inspection the district environmental officer and invited the advanced learners of the nearby school to investigate pollution concerns. The following observations were made:

(i) At the tannery: Learners observed two waste streams; one was green and the other was orange in colour. The manager explained that the colour changes with the oxidation state of chromium. He also noted that the steel was used because they withstand the high temperature better than the magnesium alloy pipes which melted during the earlier trial.

(ii) At the water treatment plant: A waste sample suspected to contain residual manganese (II) salt showed a similar behavior when treated with different alkalis but gave violet coloration when treated with a bismuthate(V) salt which identified it.

(iii) At the fertilizer factory: The chemist showed learners a green iron waste solution which turned brown and solution remained brown for days. They also noted that a

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portion the brown waste solution formed a stable solid complex when treated with excess cyanide ions compared to calcium ions previously used.

Additionally, learners were provided with the following data to better their understanding.

Substance	Mg	Ca	Cr	Mn	Fe
Melting point (^{0}C)	650	842	1907	1246	1538

The industrial chemists explained that these observations were due to the unique properties of transition elements but the learners did not understand their explanation and they have come to you for further guidance. ***(The atomic numbers of Mg, Ca, Cr, Mn and Fe are 12, 20, 24, 25 and 26 respectively.)***

Task:

As a learner of chemistry;

a) Using the data provided, analyse the observation made at the tannery.
b) (i) Explain the chemical reactions that were carried out at the water treatment plant.
   (ii) Account for the observations made at the fertilizer factory.
c) Analyse the industrial importance of chromium, manganese and iron and their impact on environment.

Item 4

Different oxides are used in industries such as cement, fertilizers, and ceramics, where their properties affect heating, safety, and environmental pollution.

Farmers near a certain factory reported crop damage and faster corrosion of iron sheets, while workers and engineers raised concerns about irritating oxide fumes and the high temperatures required to process some oxide materials.

Concerned about the increasing complaints, the district science and environment committee organized a joint industrial inspection involving chemists, environmental officers, and Senior Six chemistry learners from neighboring schools.

Samples of several oxides formed by Period 3 elements were collected from different industrial processes and analyzed in the laboratory. The learners were provided with the following information.

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Element	Na	Mg	Al	Si	P	S	Cl
Atomic number	11	12	13	14	15	16	17
Oxide	Na_2O	MgO	Al_2O_3	SiO_2	P_2O_5	SO_3	Cl_2O_7
Melting point (°C)	1275	2827	2007	1607	560	30	–91

During the inspection, the learners observed that some oxides reacted vigorously with water, some dissolved only in alkaline solutions, while others contributed to acidic emissions in the atmosphere. The industrial chemists explained that the behavior of these oxides was closely linked to their bonding, structure, and industrial applications.

The district committee later requested the learners to prepare a scientific report that could help industries improve safety, reduce pollution, and select suitable oxide materials for industrial use.

Task:

As one of the senior six learners of chemistry:

a) Explain why some oxides melt easily while others require very high temperatures during industrial processing and handling.
b) Analyse as to why some oxides reacted with water while others reacted only with sodium hydroxide during the industrial investigation.
c) Assess the environmental, health, or industrial implications associated with the production and use of the oxides investigated, and recommend suitable safety and pollution control measures for the industries.

Part II

Respond to ant one item in this part.

Item 5

A group of young entrepreneurs in Uganda intends to establish a small-scale enterprise specializing in chemical manufacturing and food processing. The business aims to manufacture and sell a variety of products including an orange azo dye for cotton fabrics, a fruity flavoring agent for beverages, a solvent for laboratory cleaning and a disinfectant.

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Prior to commencing production, the entrepreneurs need to identify suitable organic compounds and appropriate methods for their production. The compounds considered for use are **benzene** and **propanal.**

During preliminary trials, they observe that some products are highly colored while others have strong fruity smells and they wonder why. The cooperative also reacted separately benzene with fuming sulphuric acid to make a Sulphur compound and also propanal with 2,4-dinitrophenylhydrazine but lack information of how the reactions actually occur.

They also discovered that several products can be made from the two starting materials and they consider preparing an orange phenol-azo dye and **ethyl butanoate** from **benzene** and **propanal** respectively but lack expertise.

The cooperative also lacks understanding of the suitability of different organic products for specific intended applications in production of dyes and flavorings.

Task:

As a learner of chemistry;

a) Explain the differences in the physical properties of the compounds considered for use.
b) Help the cooperative solve their challenges about the reactions and formation of the intended products.
c) Assess the suitability of the azo dye and ethyl butanoate for use in the production of a specified product the company intends to make.

Item 6

In western Uganda, a cosmetics company produces body creams and perfumes but faces problems of body creams getting spoilt on exposure to moisture producing bad smells and perfumes losing their smell quickly.

Andrew, a product developer, noted that the company uses compounds **M** and **N** to make body creams and perfumes respectively. He also found out that packaging material **T** is used to store body creams and perfumes. Laboratory analysis of the body creams, perfumes and packaging material gave the following results.

(i) The structural formula of **M** is $CH_3(CH_2)_{16}COOCH_2CH_3$
(ii) The structural formula of **N** is $C_6H_5COOCH_2CH_3$
(iii) **M** is prepared by reacting a fatty acid, octadecanoic acid from palm oil with an alcohol.

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(iv) **N** can be prepared from benzene and when reacted with dimethylamine, it forms different products one of which can be used as a solvent.

(v) **T** is produced from a monomer formed when propan-2-ol reacts with concentrated sulphuric acid at 170 °C.

Andrew further requires detailed information about; the synthesis of **M, N** and **T**, formation of the functional ingredients, and the suitability and environmental impact of **M, N** and **T**.

Task:

As a learner of chemistry;

a) Explain why the body creams develop bad smells and why perfume smell does not last long.

b) Analyse the reactions leading to synthesis of;
   (i) **M** and **T**
   (ii) **N** and its reaction highlighted in the scenario.

c) Assess the environmental impact of substance **T**.

END

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